

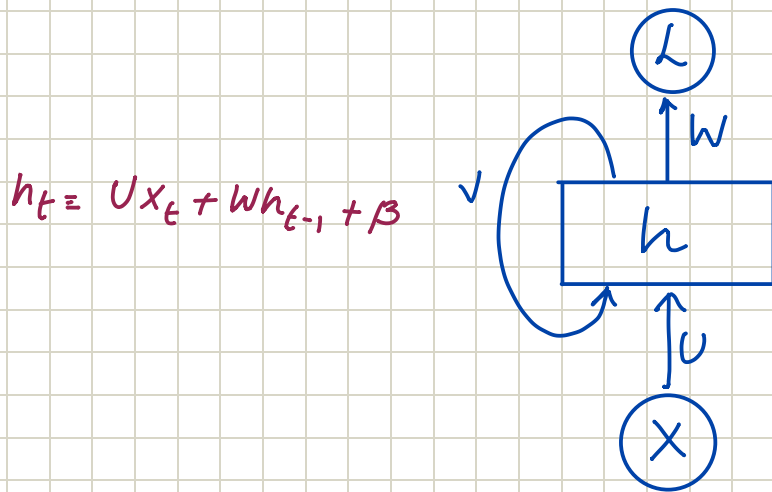
Recurrent Neural Network

Recurrent Neural Network differs from normal Neural Network in how they process the information. While standard neural network pass information in one direction, RNN feed information back into network at each step.

Mainly keeps memory of previous steps help the network understand context & make better predictions

Key Components of RNNs:

(i) Recurrent Neuron:



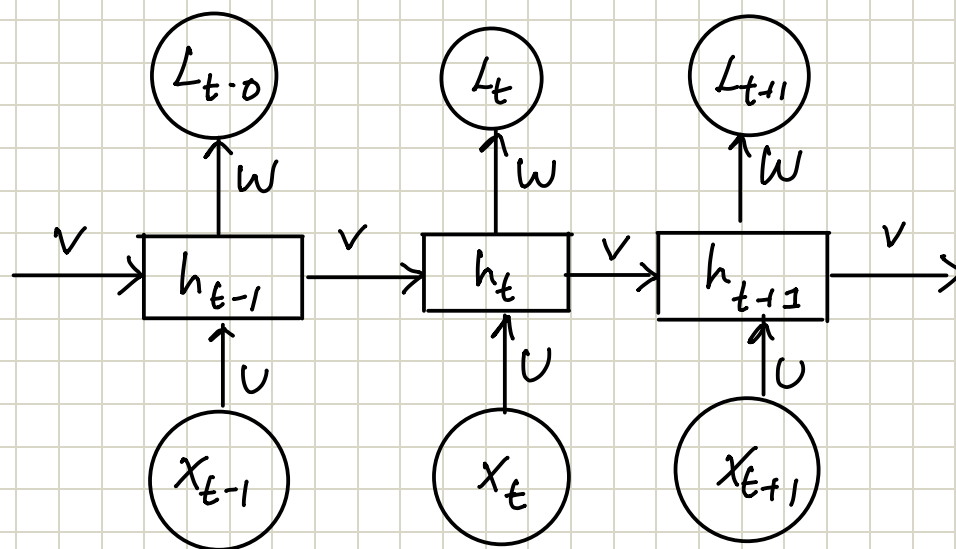
This recurrent unit hold hidden state that stores information about previous inputs in a sequence.

It can "remember" info from prior steps by feeding back their hidden state, allowing them to capture dependency across time

Recurrent Neuron
(or)

Recurrent Unit

(ii) RNN unfolding:

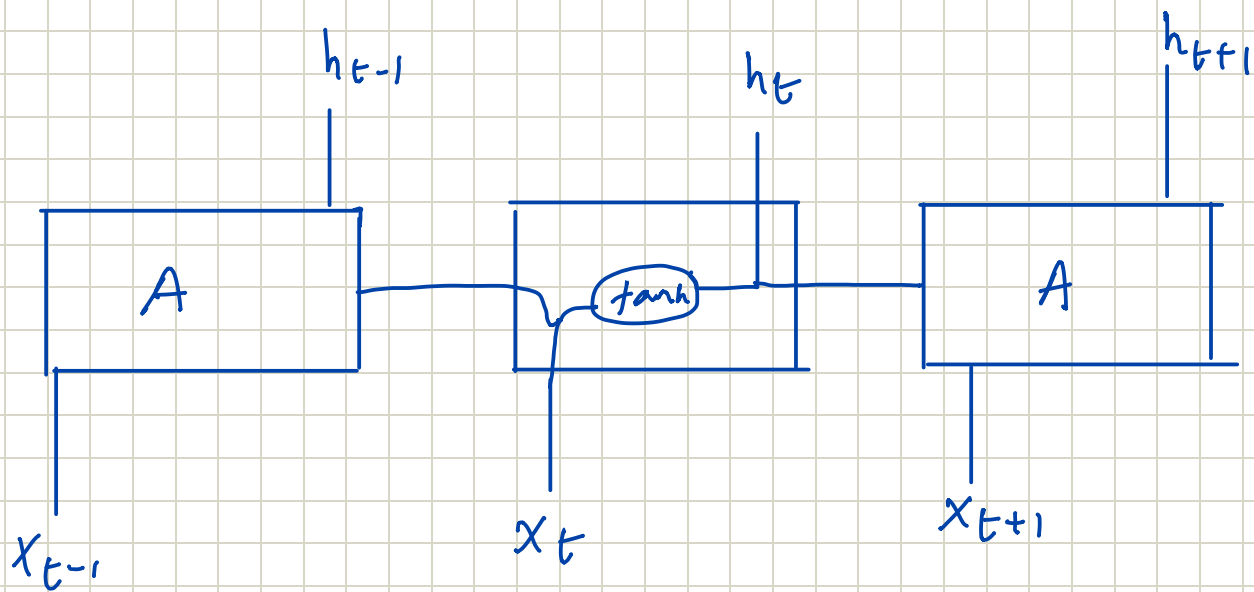


$\therefore$ Regardless how many times we unroll a RNN, the weights & biases are shared across every input.

Types of Recurrent Neural Network:

- (i) One-to-One (Image Classification)
- (ii) One to Many (Image Captioning)
- (iii) Many to One (Sentiment Analysis)
- (iv) Many to Many (Fixed Length) (Named Entity Recognition)
- (v) Many to Many (Variable length) (Machine translation)

RNN Architecture:

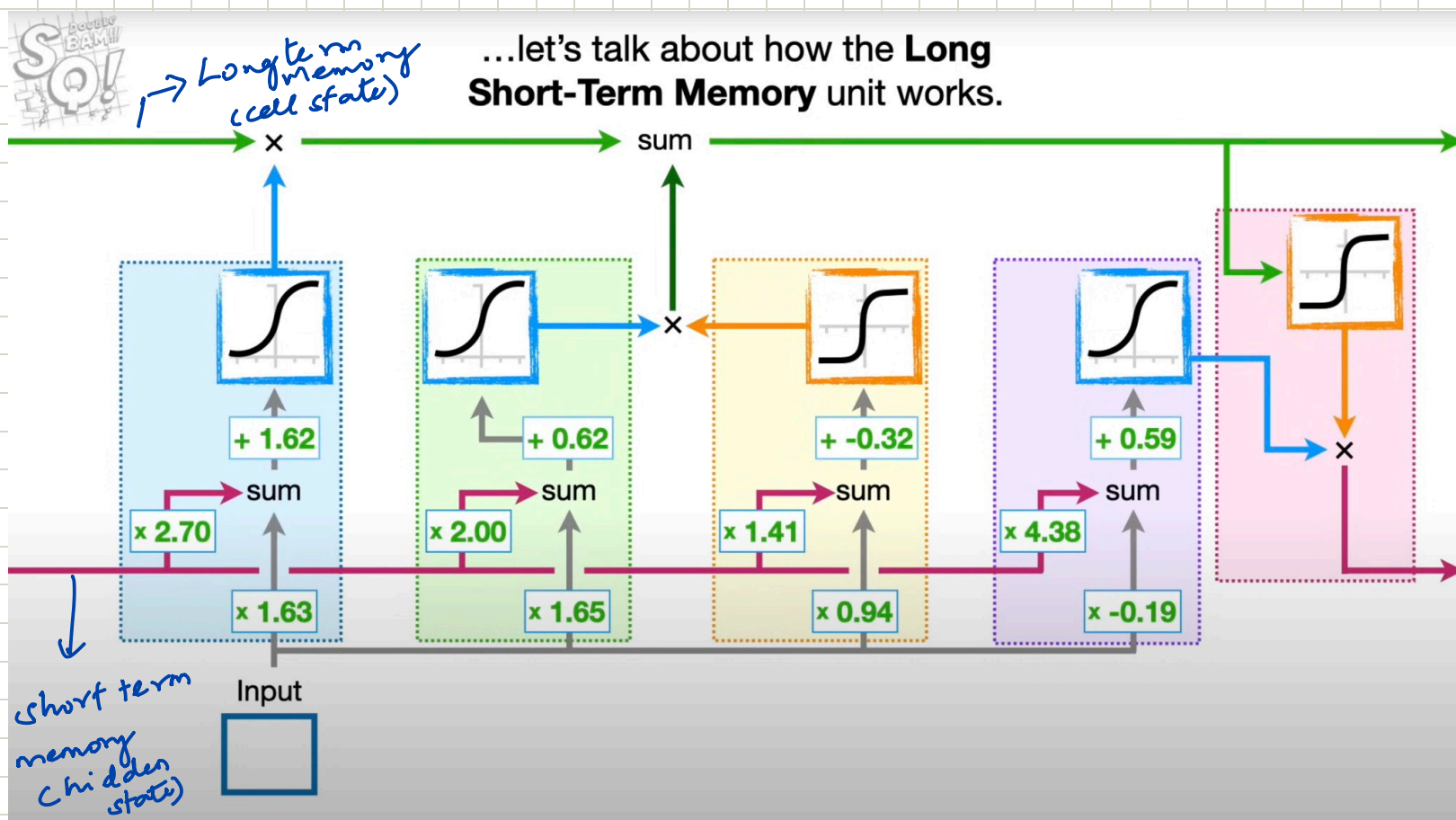


$$\text{hidden state}_{(h_t)} = \sigma(U \cdot x_t + W \cdot h_{t-1} + B)$$

Long Short-term Memory: (LSTM)

LSTM is a type of RNN that is designed to avoid exploding/vanishing gradient problem.

The main idea behind LSTM is they have two separate paths to make predictions. One is long term and other is short term.



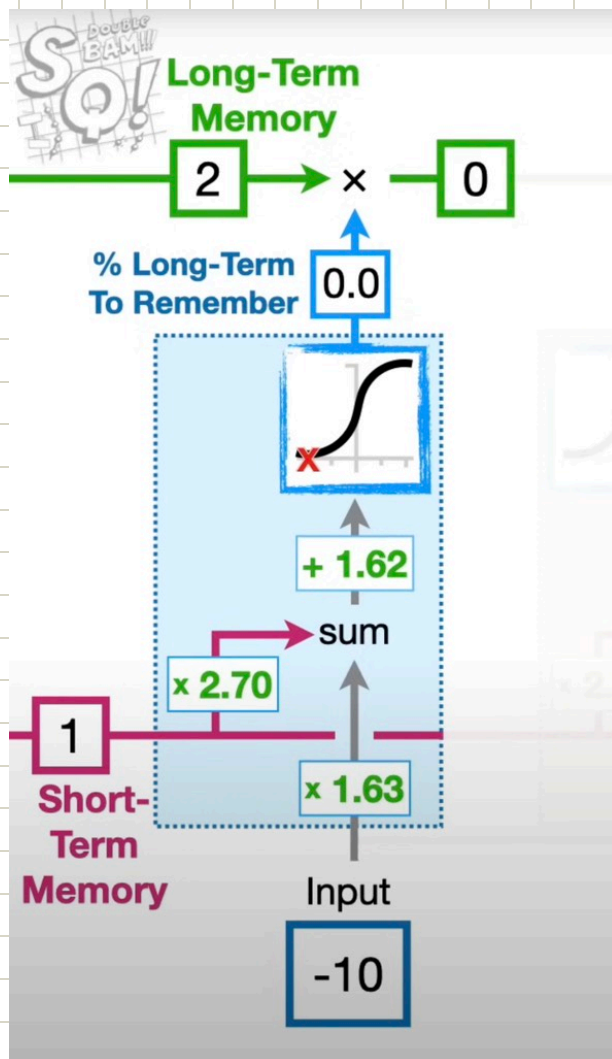
Long term Memory: It is modified only by (x) & (sum) not (cell state) influenced by weights & biases. So No gradient problem.

Short term Memory: It is directly connected to weights that can modify them.
(Hidden state)

First Stage: (The percent to remember)

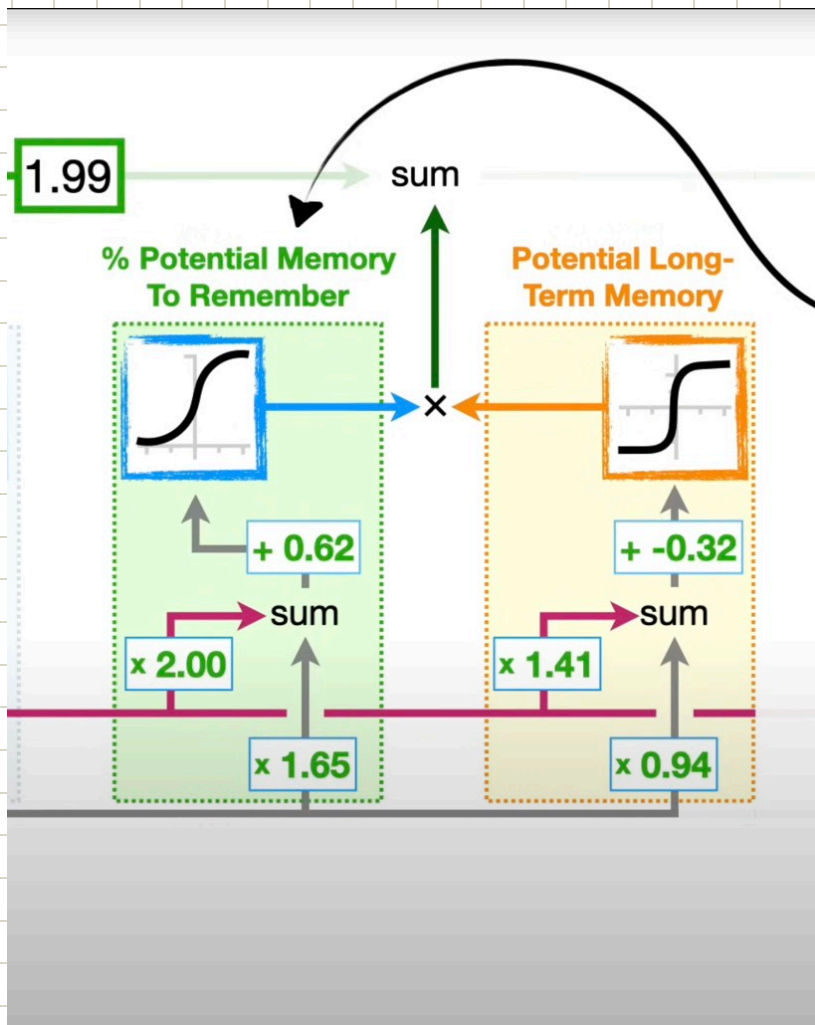
Forget Gate: determines the percentage of Long term memory should be remembered.

Forget Gate



2nd Stage: (Update the Longterm Memory)

Input Gate:



From short term Memory we first calculate potential long term Memory using input, short term memory (hidden state) and tanh functions. Later calculate how much percentage of this potential Long term memory to remember & then add it to cell state (Long term Memory).

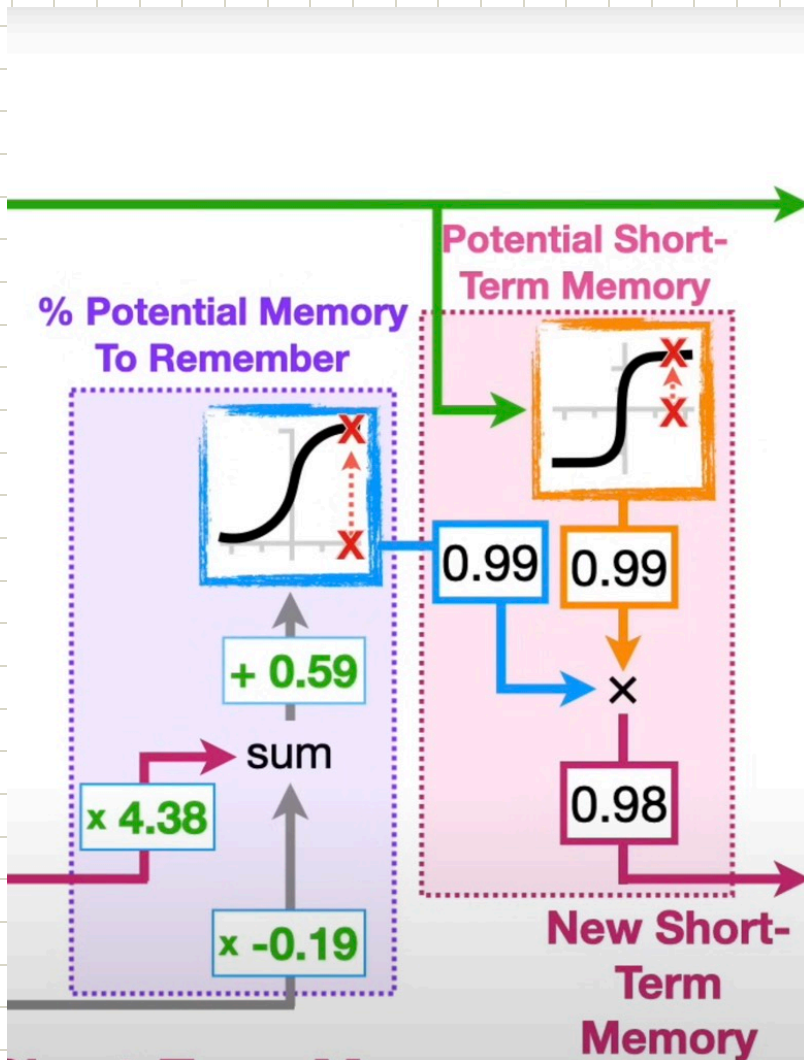
So far:

Percent to Remember (Stage 1) → how much to update Long term Memory from input & short term (Stage 2)

3rd Stage: Update the short term Memory:

Output Gate:

Here once we have new Long term Memory, we use that to calculate the potential short term Memory using "tanh" and also using input, short term Memory and sigmoid to calculate how much percentage of potential short term memory to pass on. This new short term memory is the output of the entire LSTM unit.



Note: Watch StatQuest for any Queries.

